Project 2 – TMA4300

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## Warning: package 'glmmTMB' was built under R version 4.4.3
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Problem 1

We suppose that the joint density of U, Z up to a normalizing constant is

$$f^*_{U,Z}(u,z) = \begin{cases} 1, & \text{for } 0 < u < e^{-z^2/2}, \\ 0, & \text{elsewhere.} \end{cases}$$

a)

If a joint density is known up to proportionality, the full conditional of one variable is obtained by

$$f_{X|Y}(x|y) \propto f^*_{U,Z}(u,z) \textit{as a function of } x$$

Conditional ditribution of $U|Z = z$

Fix z. The joint density becomes

$$f^*_{U,Z}(u,z) = \begin{cases} 1, & \text{for } 0 < u < e^{-z^2/2}, \\ 0, & \text{elsewhere.} \end{cases}$$

As a function of u, this is constant on the interval $(0, e^{-z^2/2})$.

The normalizing constant is

$$\int_0^{e^{-z^2/2}} 1\, du = e^{-z^2/2}.$$

Therefore,

$$f_{U|Z}(u \mid z) = \frac{1}{e^{-z^2/2}} = e^{z^2/2}, \quad 0 < u < e^{-z^2/2}.$$

Hence,

$$U \mid Z = z \sim \text{Unif}\big(0, e^{-z^2/2}\big).$$

.

Conditional ditribution of $Z|U = u$

Fix u. The support condition is $0 < u < e^{-z^2/2}$.

Solving for z:

$$u < e^{-z^2/2} \quad \Longleftrightarrow \quad \log u < -\frac{z^2}{2}$$

$$-2\log u > z^2$$

$$|z| < \sqrt{-2\log u}$$

Define

$$a(u) = \sqrt{-2\log u}$$

Therefore, the support of the conditional distribution is

$$z \in (-a(u), a(u)).$$

Within this interval the joint density is constant and equal to 1. The normalizing constant is therefore the length of the interval:

$$\int_{-a(u)}^{a(u)} 1\, dz = 2a(u).$$

Thus, the full conditional density of $Z \mid U = u$ is

$$f_{Z|U}(z \mid u) = \frac{1}{2a(u)}, \quad |z| < a(u),$$

where

$$a(u) = \sqrt{-2\log u}.$$

Thus,

$$Z \mid U = u \sim \mathrm{Unif}\big(-a(u), a(u)\big).$$

Gibbs Sampler

Given an initial valid point $(U^{(0)}, Z^{(0)})$ satisfying $0 < U^{(0)} < e^{-(Z^{(0)})^2/2}$, iterate for $t = 1, \ldots, n$:

1. Sample
$$U^{(t)} \sim \mathrm{Unif}\left(0, e^{-(Z^{(t-1)})^2/2}\right).$$

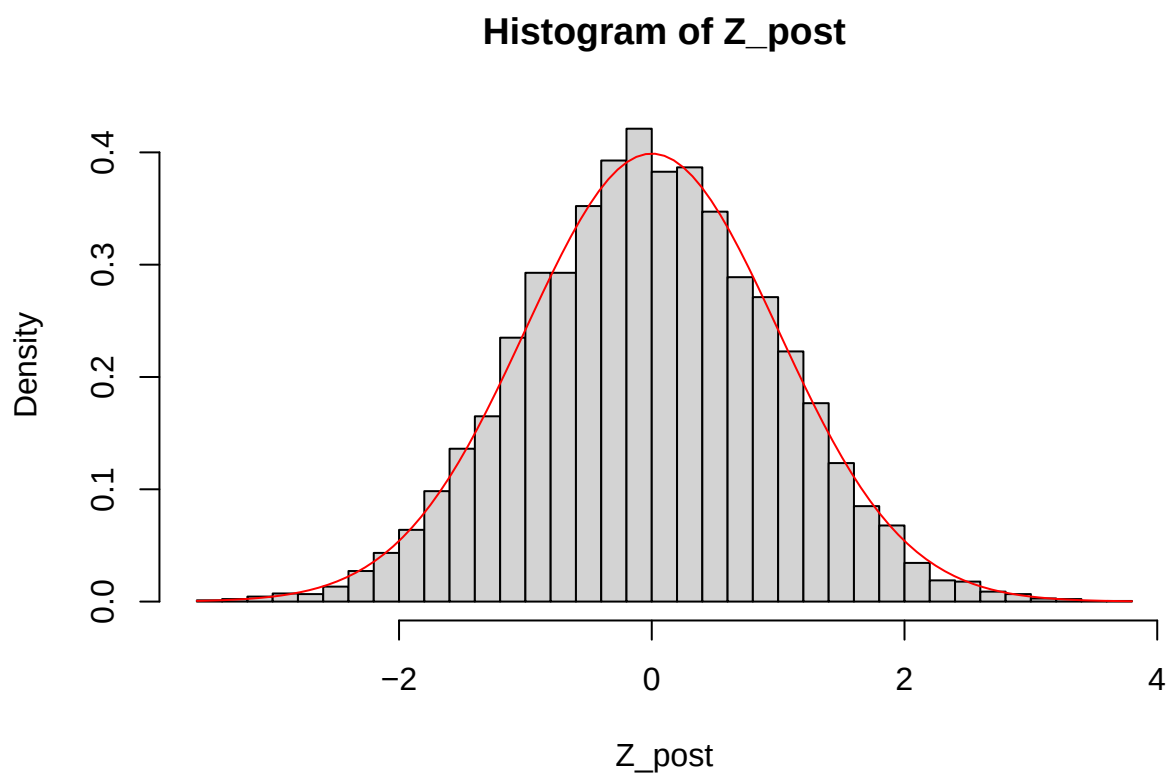
2. Define
$$a^{(t)} = \sqrt{-2\log U^{(t)}}.$$

3. Sample
$$Z^{(t)} \sim \mathrm{Unif}\left(-a^{(t)},\, a^{(t)}\right).$$

This iterative scheme defines a Markov chain whose stationary distribution is the joint density $f_{U,Z}(u, z)$.The marginal stationary distribution of Z is $\mathcal{N}(0, 1)$.

```
## [1] -0.01857042
```

```
## [1] 0.9699554
```



The posterior sample mean is approximately

$$\hat{\mu} \approx -0.019,$$

and the empirical variance is

$$\hat{\sigma}^2 \approx 0.97.$$

Both values are close to the theoretical mean and variance,

$$\mathbb{E}[Z] = 0, \qquad \mathrm{Var}(Z) = 1,$$

of a standard normal distribution.

The histogram aligns well with the standard normal density curve, confirming that the Gibbs sampler correctly targets

$$Z \sim \mathcal{N}(0, 1).$$

b)

We determine the marginal distribution of Z by integrating out U.

Recall that the joint density (up to normalization) is

$$f^*_{U,Z}(u,z) = \begin{cases} 1, & 0 < u < e^{-z^2/2}, \\ 0, & \text{otherwise.} \end{cases}$$

The marginal density of Z is therefore

$$f_Z(z) = \int_0^{e^{-z^2/2}} 1\, du = e^{-z^2/2}.$$

Thus,

$$f_Z(z) \propto e^{-z^2/2}.$$

Since

$$\int_{-\infty}^{\infty} e^{-z^2/2}\, dz = \sqrt{2\pi},$$

the normalized density is

$$f_Z(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}.$$

Therefore,

$$Z \sim \mathcal{N}(0,1).$$

Hence, discarding the U-values from the Gibbs sampler yields a sample from a standard normal distribution.

c)

Correlation between successive Gibbs samples Z_{n-1} and Z_n}

From the Gibbs sampler we have

$$Z_n \mid U_n \sim \text{Unif}\big(-a(U_n), a(U_n)\big), \qquad a(u) = \sqrt{-2\log u}.$$

This conditional distribution is symmetric around zero. Therefore,

$$\mathbb{E}[Z_n \mid U_n] = 0.$$

By the law of total expectation,

$$\mathbb{E}[Z_n] = \mathbb{E}[\mathbb{E}[Z_n \mid U_n]] = \mathbb{E}[0] = 0.$$

Since the Gibbs sampler has stationary distribution $Z_n \sim \mathcal{N}(0,1)$, it follows that

$$\mathbb{E}[Z_{n-1}] = \mathbb{E}[Z_n] = 0.$$

We now compute the covariance. By definition,

$$\text{Cov}(Z_{n-1}, Z_n) = \mathbb{E}[Z_{n-1}Z_n] - \mathbb{E}[Z_{n-1}]\mathbb{E}[Z_n].$$

Because both expectations are zero, this reduces to

$$\text{Cov}(Z_{n-1}, Z_n) = \mathbb{E}[Z_{n-1}Z_n].$$

Using the tower property (law of iterated expectation),

$$\mathbb{E}[Z_{n-1}Z_n] = \mathbb{E}[\mathbb{E}[Z_{n-1}Z_n \mid U_n]]\,.$$

In the Gibbs sampler, Z_n is generated from U_n independently of Z_{n-1}. Hence, conditional on U_n, the variables Z_{n-1} and Z_n are independent. Therefore,

$$\mathbb{E}[Z_{n-1}Z_n \mid U_n] = Z_{n-1}\,\mathbb{E}[Z_n \mid U_n].$$

Since $\mathbb{E}[Z_n \mid U_n] = 0$, we obtain

$$\mathbb{E}[Z_{n-1}Z_n \mid U_n] = 0.$$

Taking expectations again,

$$\mathbb{E}[Z_{n-1}Z_n] = 0.$$

Thus,

$$\mathrm{Cov}(Z_{n-1}, Z_n) = 0.$$

Successive Gibbs samples Z_{n-1} and Z_n are uncorrelated in stationarity. However, they are not independent, since both depend on the intermediate variable U_n.

Correlation between successive second moments Z_{n-1}^2 and Z_n^2

Recall that

$$Z_n \mid U_n \sim \mathrm{Unif}\big(-a(U_n), a(U_n)\big), \qquad a(u) = \sqrt{-2\log u}.$$

For a symmetric uniform distribution on $(-a, a)$ we have

$$\mathbb{E}[X^2] = \frac{a^2}{3}.$$

Therefore,

$$\mathbb{E}[Z_n^2 \mid U_n] = \frac{a(U_n)^2}{3} = \frac{-2\log U_n}{3}.$$

Unlike the first moment case, this conditional second moment depends explicitly on U_n.

We now compute the covariance using the law of total covariance:

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2) = \mathbb{E}\big[\mathrm{Cov}(Z_{n-1}^2, Z_n^2 \mid U_n)\big] + \mathrm{Cov}\big(\mathbb{E}[Z_{n-1}^2 \mid U_n], \mathbb{E}[Z_n^2 \mid U_n]\big)\,.$$

In the Gibbs sampler, Z_n is generated from U_n independently of Z_{n-1}. Hence, conditional on U_n, the variables Z_{n-1} and Z_n are independent, and therefore

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2 \mid U_n) = 0.$$

Thus,

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2) = \mathrm{Cov}\big(\mathbb{E}[Z_{n-1}^2 \mid U_n], \mathbb{E}[Z_n^2 \mid U_n]\big)\,.$$

From the previous calculation,

$$\mathbb{E}[Z_{n-1}^2 \mid U_n] = \mathbb{E}[Z_n^2 \mid U_n] = \frac{-2\log U_n}{3}.$$

Therefore,

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2) = \mathrm{Var}\left(\frac{-2\log U_n}{3}\right).$$

To evaluate this variance, we determine the marginal distribution of U_n in stationarity.
It can be shown that

$$T = -2\log U_n \sim \chi_3^2.$$

Hence,

$$\mathbb{E}[T] = 3, \qquad \mathrm{Var}(T) = 6.$$

It follows that

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2) = \mathrm{Var}(T/3) = \frac{6}{9} = \frac{2}{3}.$$

Successive second moments are positively correlated:

$$\mathrm{Cov}(Z_{n-1}^2, Z_n^2) = \frac{2}{3} \neq 0.$$

Thus, although successive Gibbs samples Z_{n-1} and Z_n are uncorrelated, their squared values exhibit positive correlation due to the shared dependence on the intermediate variable U_n.

Problem 2

a)

```
##  Family: gaussian  ( identity )
## Formula:          y ~ (1 | i)
## Data: data
##
##       AIC       BIC    logLik -2*log(L)  df.resid
##     280.3     288.1    -137.2     274.3        97
##
## Random effects:
##
## Conditional model:
##  Groups   Name        Variance Std.Dev.
##  i        (Intercept) 0.2301   0.4797
##  Residual             0.7560   0.8695
## Number of obs: 100, groups:  i, 20
##
## Dispersion estimate for gaussian family (sigma^2): 0.756
##
## Conditional model:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  10.1888     0.1381   73.79   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The maximum likelihood estimate of the overall mean is $\hat{\mu} = 10.1888$. The estimated variance of the random intercepts is $\widehat{\mathrm{Var}}(\gamma_i) = 0.2301$, corresponding to $1/\nu$, and the estimated residual variance is $\widehat{\mathrm{Var}}(\epsilon_{ij}) = 0.7560$, corresponding to $1/\tau$. These estimates will later be used to check that the Gibbs sampler implementation produces reasonable results.

b) Full conditional distributions

From the hierarchical model

$$y_{ij} \mid \mu, \gamma_i, \tau \sim N(\mu + \gamma_i, \tau^{-1}),$$

$$\gamma_i \mid \nu \sim N(0, \nu^{-1}),$$

with priors

$$\pi(\mu) \propto 1, \qquad \tau \sim \text{Gamma}(\alpha, \alpha), \qquad \nu \sim \text{Gamma}(\alpha, \alpha),$$

we derive the following full conditional distributions.
Let $N = \sum_{i=1}^{m} n_i$.

Full conditional of μ Combining the likelihood with the flat prior for μ, we obtain

$$\mu \mid \gamma, \tau, y \sim N(\bar{y}_\gamma, (N\tau)^{-1}),$$

where

$$\bar{y}_\gamma = \frac{1}{N} \sum_{i=1}^{m} \sum_{j=1}^{n_i} (y_{ij} - \gamma_i).$$

Full conditional of γ_i Using the likelihood contribution from cluster i together with the prior $\gamma_i \sim N(0, \nu^{-1})$, we obtain

$$\gamma_i \mid \mu, \tau, \nu, y \sim N(m_i, (\nu + n_i\tau)^{-1}),$$

where

$$m_i = \frac{\tau \sum_{j=1}^{n_i} (y_{ij} - \mu)}{\nu + n_i\tau}.$$

Full conditional of τ Let

$$S = \sum_{i=1}^{m} \sum_{j=1}^{n_i} (y_{ij} - \mu - \gamma_i)^2.$$

Combining the Gaussian likelihood with the prior $\tau \sim \text{Gamma}(\alpha, \alpha)$ gives

$$\tau \mid \mu, \gamma, \nu, y \sim \text{Gamma}\left(\alpha + \frac{N}{2}, \alpha + \frac{S}{2}\right).$$

Full conditional of ν Using the prior for γ_i and $\nu \sim \text{Gamma}(\alpha, \alpha)$, we obtain

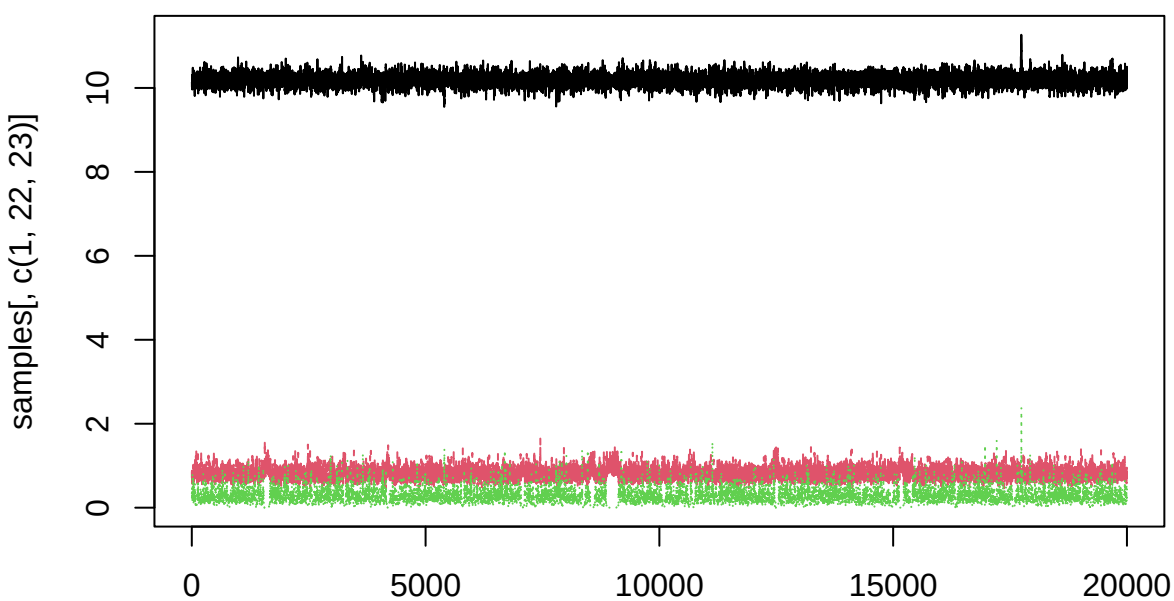
$$\nu \mid \mu, \gamma, \tau, y \sim \text{Gamma}\left(\alpha + \frac{m}{2},\, \alpha + \frac{1}{2}\sum_{i=1}^{m} \gamma_i^2\right).$$

c) Implementation of a Gibbs sampler

```
##  [1] 10.186949480 -0.228328265 -0.241998135 -0.308130141  0.445724338
##  [6]  0.001276209 -0.195878016  0.081798904  0.209191417  0.310980433
## [11]  0.052458426  0.112042491 -0.069377360 -0.319505738 -0.513020222
## [16]  0.656524132 -0.441780123 -0.323819534  0.752023533  0.063353793
## [21] -0.013827620  0.802720740  0.246525959
```

The Gibbs sampler was run for 20000 iterations to sample from the joint posterior distribution of the model parameters. The posterior mean of the global intercept is approximately $\mu \approx 10.19$, which is very close to the maximum likelihood estimate obtained in part (a). The posterior mean of the residual variance is approximately $1/\tau \approx 0.80$ and the posterior mean of the random intercept variance is approximately $1/\nu \approx 0.25$. These values are close to the corresponding maximum likelihood estimates (0.756 and 0.2301), suggesting that the Gibbs sampler produces reasonable estimates and appears to converge to the correct posterior distribution.

d)



The trace plots for the parameters appear stable and fluctuate around constant levels, indicating that the Gibbs sampler mixes reasonably well and that the Markov chains have reached a stationary distribution. The chain for the global intercept μ is centered around approximately 10.2, while the residual variance $1/\tau$ fluctuates around about 0.8 and the random intercept variance $1/\nu$ around about 0.25. These values are consistent with the maximum likelihood estimates obtained in part (a). The trace plot for $1/\nu$ shows slightly larger variability and occasional spikes compared to the other parameters, which is expected for variance parameters with weakly informative priors, but overall the chain still appears stable.

Problem 4

We consider the state equation

$$u_t = u_{t-1} + r\left(1 - \left(\frac{e^{u_{t-1}}}{K}\right)^{\theta}\right) + e_t, \qquad e_t \sim \mathcal{N}(0, \sigma), \tag{1}$$

and the observation equation

$$y_t \mid u_t \sim \text{Poisson}(ce^{u_t}). \tag{2}$$

a)

To show that the model is non-identifiable, it suffices to demonstrate that different parameter values yield the same distribution for the observations $y = (y_1, \ldots, y_n)$.

Let $a \in \mathbb{R}$ and define

$$u_t^* = u_t + a, \qquad K^* = Ke^a, \qquad c^* = ce^{-a},$$

with

$$r^* = r, \qquad \theta^* = \theta, \qquad \sigma^* = \sigma.$$

Then

$$\frac{e^{u_{t-1}^*}}{K^*} = \frac{e^{u_{t-1}+a}}{Ke^a} = \frac{e^{u_{t-1}}}{K},$$

so the state equation remains unchanged:

$$u_t^* = u_{t-1}^* + r\left(1 - \left(\frac{e^{u_{t-1}^*}}{K^*}\right)^{\theta}\right) + e_t.$$

For the observation model,

$$c^* e^{u_t^*} = (ce^{-a})e^{u_t+a} = ce^{u_t},$$

so the conditional distribution of y_t is unchanged.

Since u_1 has an improper uniform distribution on $\mathbb{R}$, it is invariant under shifts. Hence, the transformation

$$(u_t, K, c) \mapsto (u_t + a, Ke^a, ce^{-a})$$

leaves the likelihood unchanged, and the model is non-identifiable.

Identifiability. The non-identifiability arises from a shift invariance in u_t. The quantity

$$\log K + \log c$$

is invariant under the transformation, implying that only the product cK is identifiable. To remove this invariance, we impose the constraint

$$c = 1,$$

which yields an identifiable model without altering its structure.

b)

To resolve the non-identifiability shown in part (a), we fix $c = 1$. The parameters r, K, θ, σ are estimated, with K, θ, and σ optimized on the log-scale to enforce positivity.

We write the negative joint log-likelihood as

$$-\log f(y, u \mid \theta) = -\sum_{t=1}^{n} \log p(y_t \mid u_t) - \sum_{t=2}^{n} \log p(u_t \mid u_{t-1}),$$

where

Observation model

$$y_t \mid u_t \sim \text{Poisson}(e^{u_t}),$$

so

$$-\log p(y_t \mid u_t) = e^{u_t} - y_t u_t + \log(y_t!).$$

State model

$$u_t \mid u_{t-1} \sim \mathcal{N}(\mu_t, \sigma^2), \quad \mu_t = u_{t-1} + r\left(1 - \left(\frac{e^{u_{t-1}}}{K}\right)^{\theta}\right),$$

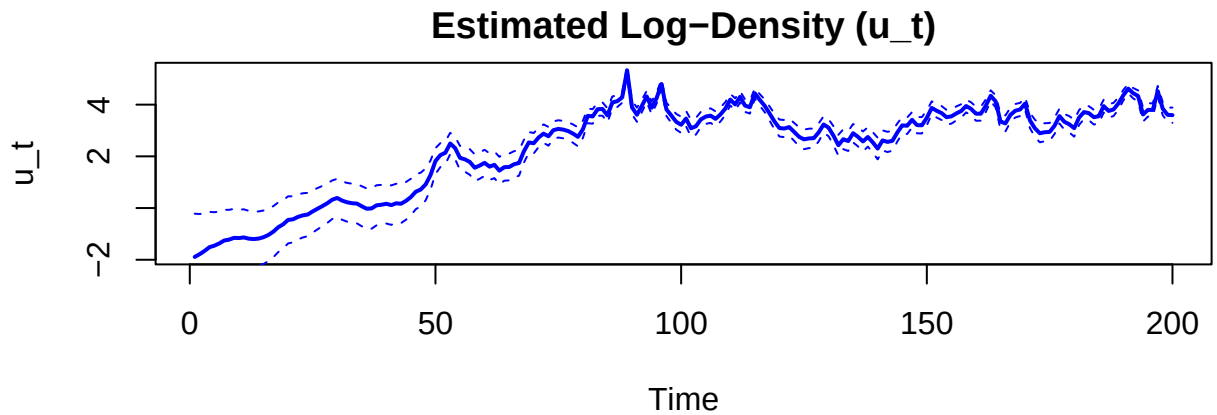
giving

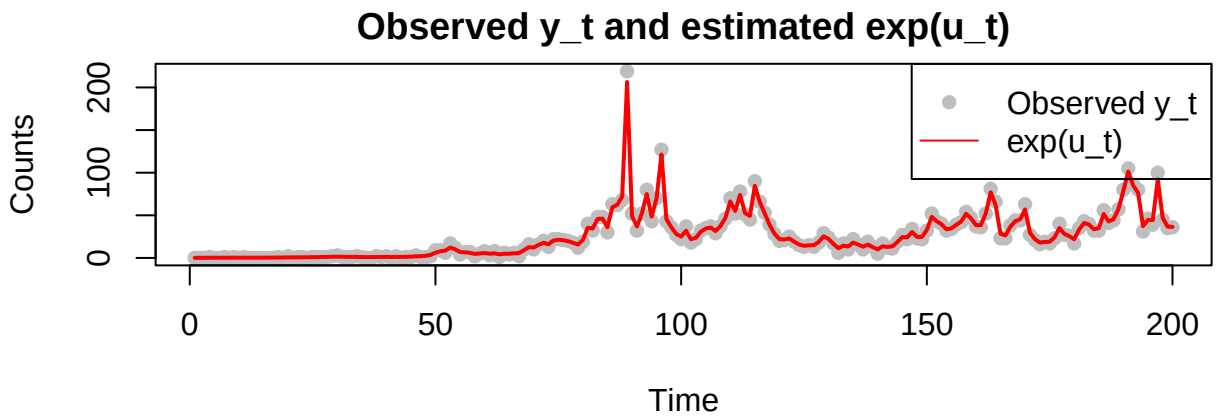
$$-\log p(u_t \mid u_{t-1}) = \frac{(u_t - \mu_t)^2}{2\sigma^2} + \log \sigma + \text{const}.$$

The latent states $u = (u_1, \ldots, u_n)$ are treated as random effects in `RTMB`. The marginal likelihood is approximated using the Laplace approximation via `MakeADFun` with `random = "u"`. Parameters K, θ, and σ are optimized on the log-scale to enforce positivity, and $c = 1$ is fixed to ensure identifiability.

c)

The marginal likelihood was maximized using the `nlminb` optimizer. The optimisation converged successfully, with a small gradient norm at the optimum, indicating a stable solution.





The estimated latent states correspond to the conditional modes obtained from the Laplace approximation. Since $c = 1$, the observation mean is $E[y_t \mid u_t] = e^{u_t}$. The fitted trajectory captures the overall trend of the data while smoothing out the Poisson noise, reflecting the underlying process dynamics.

d)

To evaluate the performance of the method, we simulated a dataset of $n = 200$ observations using known parameter values and fitted the model to the simulated data.

Results

Table 1: Comparison of true parameter values and estimates based on simulated data.

	Parameter	True value	Estimate	Std. error	95% CI
r	r	0.1	0.1134	0.0431	[0.029, 0.198]
K	K	45.0	43.2630	4.8223	[33.811, 52.715]
theta	θ	2.0	2.1603	0.7618	[0.667, 3.654]
sigma	σ	0.3	0.2718	0.0211	[0.23, 0.313]

The parameter estimates are close to the true values, and all true parameters lie within the corresponding 95% confidence intervals. This indicates that the estimation procedure performs well and that the Laplace approximation provides reliable inference for this model.

The parameter θ exhibits the largest uncertainty, reflected in a relatively wide confidence interval. This is expected, as θ governs the nonlinear density dependence and enters the model in a highly nonlinear manner, making it more difficult to estimate from finite data. In contrast, σ is estimated with high precision, indicating that the noise level is well identified. The parameters r and K are also reasonably well estimated, suggesting that the overall growth dynamics are captured effectively.

The results are based on a single simulated dataset, and the estimation performance may vary across different realizations. In addition, the Laplace approximation relies on a Gaussian approximation of the latent posterior distribution, which may be less accurate in highly nonlinear settings or for small sample sizes. Nevertheless, the results suggest that the approximation performs well in this setting.

Fixing $c = 1$ was essential to ensure identifiability. Without this constraint, the model exhibits a shift invariance in the latent states, making it impossible to uniquely determine the parameters. Imposing this constraint leads to a well-defined estimation problem and stable numerical optimization.

Standard errors

The standard errors reported by `sdreport` are based on asymptotic likelihood theory. The covariance matrix is approximated by the inverse Hessian of the Laplace-approximated marginal negative log-likelihood,

$$H(\hat{\theta})^{-1}.$$

Since parameters are optimized on the log-scale, the Delta method is used to obtain standard errors on the original scale:

$$\mathrm{Var}(g(\theta)) \approx \nabla g(\theta)^T H(\theta)^{-1} \nabla g(\theta).$$

Overall, the standard errors appear to be of a reasonable magnitude, and the agreement between estimated and true parameter values provides empirical support for the reliability of the Laplace approximation in this model.